

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Static · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/Alex/Static · 30 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/Georgia/Static · 30 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Static · 31 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Static · 30 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static · 32 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/StephenV2/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/Mansh/Static · 31 trials
Total training paths	10 · 304 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Josh/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=486 · test=61

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (RBF)	1.0000	0.0000	1.0000
SVM (Linear)	1.0000	0.0000	1.0000
Logistic Regression	0.9958	0.0125	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000
Stacking (SVM+RF+KNN+LogReg)	1.0000	0.0000	1.0000
Random Forest	0.9979	0.0063	0.9672
KNN	0.9794	0.0227	0.9672
LDA	0.9114	0.0509	0.4590

Per-class hold-out F1

Class	SVM (RBF)	SVM (Linear)	Logistic Regression	Voting (soft)	Stacking	Random Forest	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	0.952	1.000	0.471	11
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	0.889	0.375	10
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	0.909	0.348	10
Double_Phone	1.000	1.000	1.000	1.000	1.000	0.900	1.000	0.667	10
Double_Pistol	1.000	1.000	1.000	1.000	1.000	0.952	1.000	0.267	10
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.500	10
macro avg	1.000	1.000	1.000	1.000	1.000	0.967	0.966	0.438	61

External test (NewTestData)

Algorithm	External test acc
Random Forest	1.0000
Logistic Regression	0.9667

Voting Soft (SVM+RF+KNN+LogReg)	0.9667
Stacking (SVM+RF+KNN+LogReg)	0.9667
SVM (RBF)	0.9333
SVM (Linear)	0.9333
KNN	0.8000
LDA	0.3000